IBM Data Analyst Captsone Project

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EXECUTIVE SUMMARY

- Analyzes tech skill trends from surveys and job data
- JavaScript leads; Python and TypeScript are rising
- PostgreSQL is the top database choice
- React and Flask influence language use
- No-code tools shape future demand
- Insights visualized with IBM Cognos dashboards

INTRODUCTION

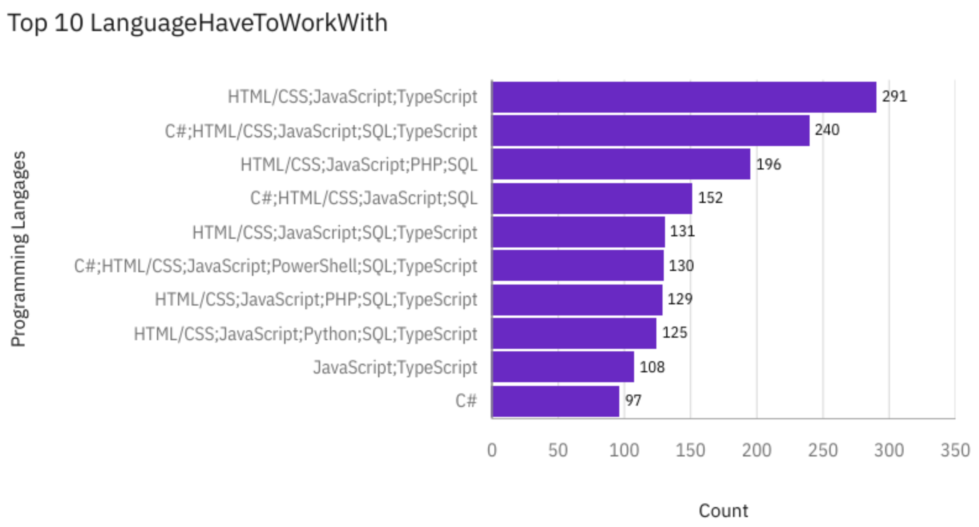
The tech landscape is rapidly evolving, and organizations must understand skill demand trends to remain competitive. This report aims to identify those trends using real-world data. It is intended for recruiters, hiring managers, and educational planners seeking to align with current and future skill requirements. The report begins with data analysis and concludes with practical implications.

METHODOLOGY

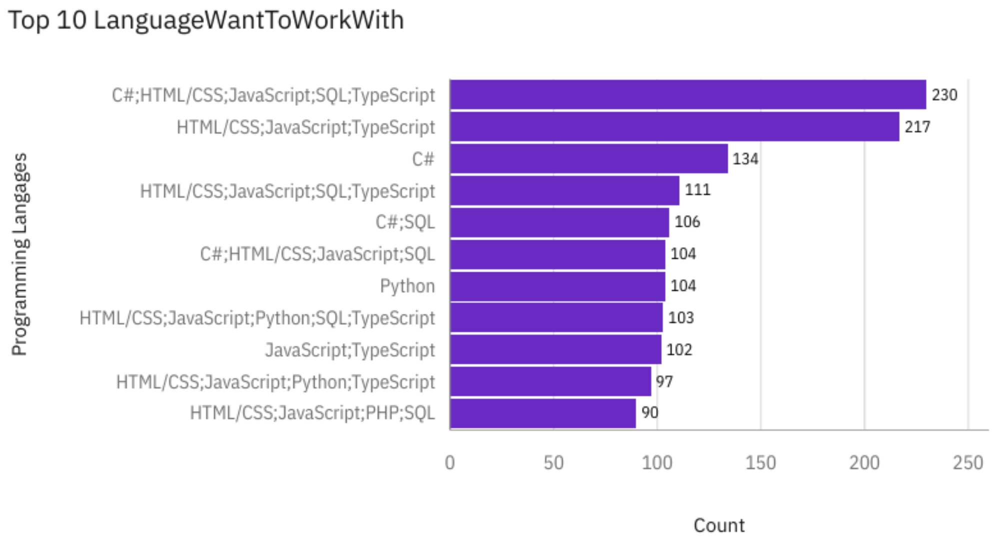
- **Data Source:** API of job postings and the Stack Overflow Developer Survey.
- **Collection Method:** JSON data extracted via scripts.
- **Data Wrangling:**
  - Duplicate removal and missing value imputation
  - Normalization of skills and categories
  - Structuring of unstructured fields (e.g., "Key Skills")
- **EDA:**
  - Analyzing value distributions, outlier handling, correlation analysis
- **Output:** A structured CSV used for dashboarding in IBM Cognos Analytics. Let’s now examine the results

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year

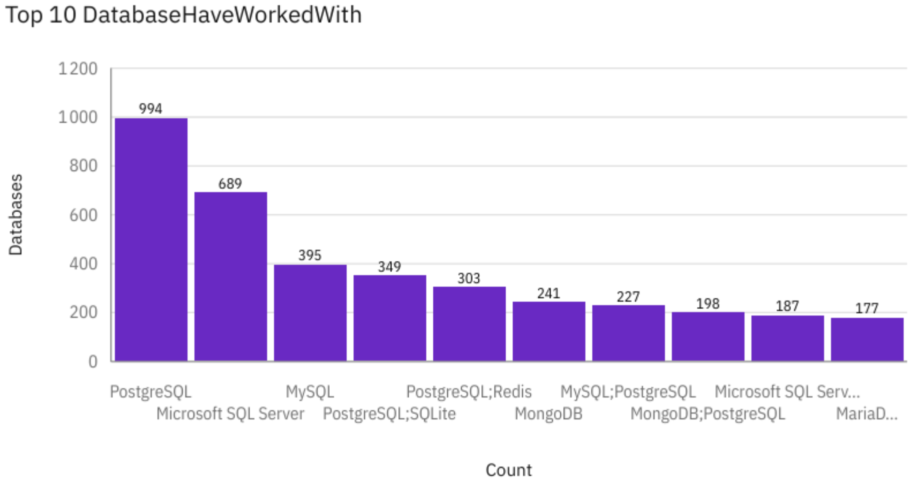


Programming Language Trends Current Usage: JavaScript, Python, Java, C#, TypeScript, SQL, etc.
Future Intent: Python and TypeScript show increased interest.

-> Organizations should pivot to versatile languages like Python and support TypeScript for web-scale applications.

DATABASE TRENDS

Current Year



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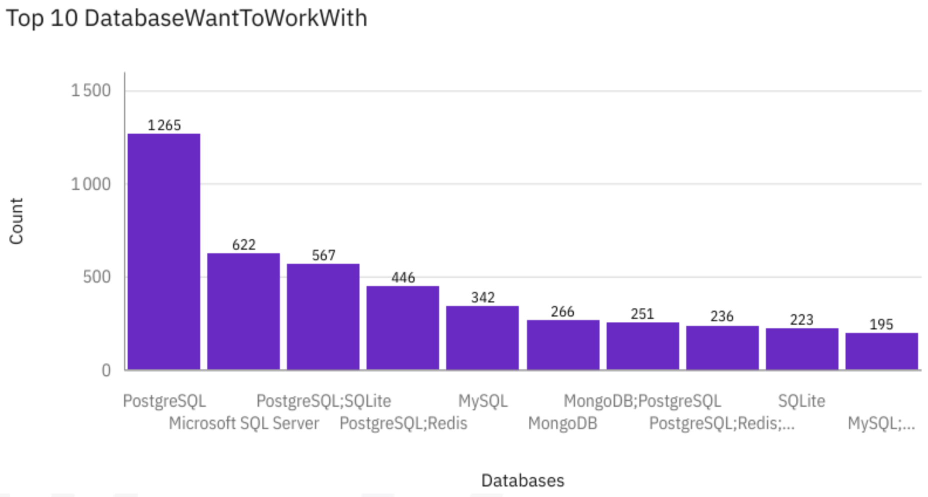
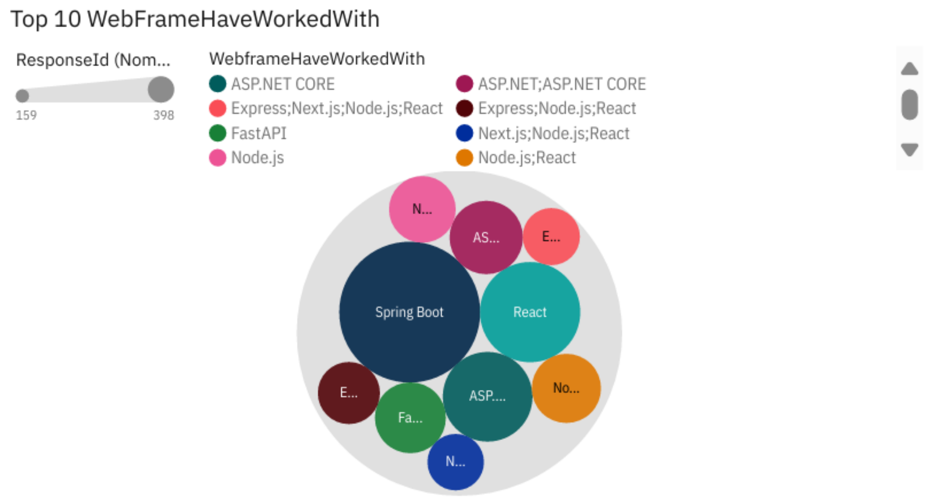


Database Trends Current: PostgreSQL and MySQL most used; others include SQLite, MongoDB.
Future: Preference for PostgreSQL remains, but Redis gain traction.

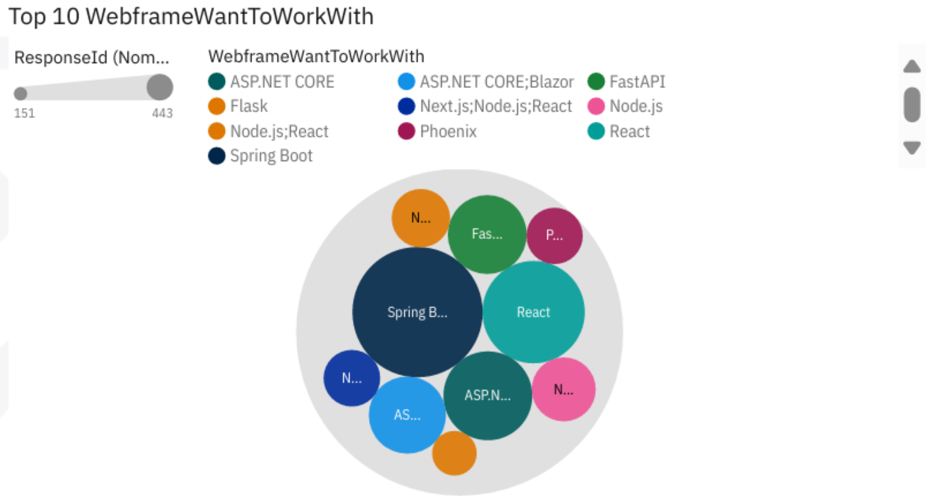
-> Cloud-native and flexible DBs will dominate; training should include NoSQL models.

WEB FRAMEWORKS AND LANGUAGE INTERPLAY

Current Year



Next Year



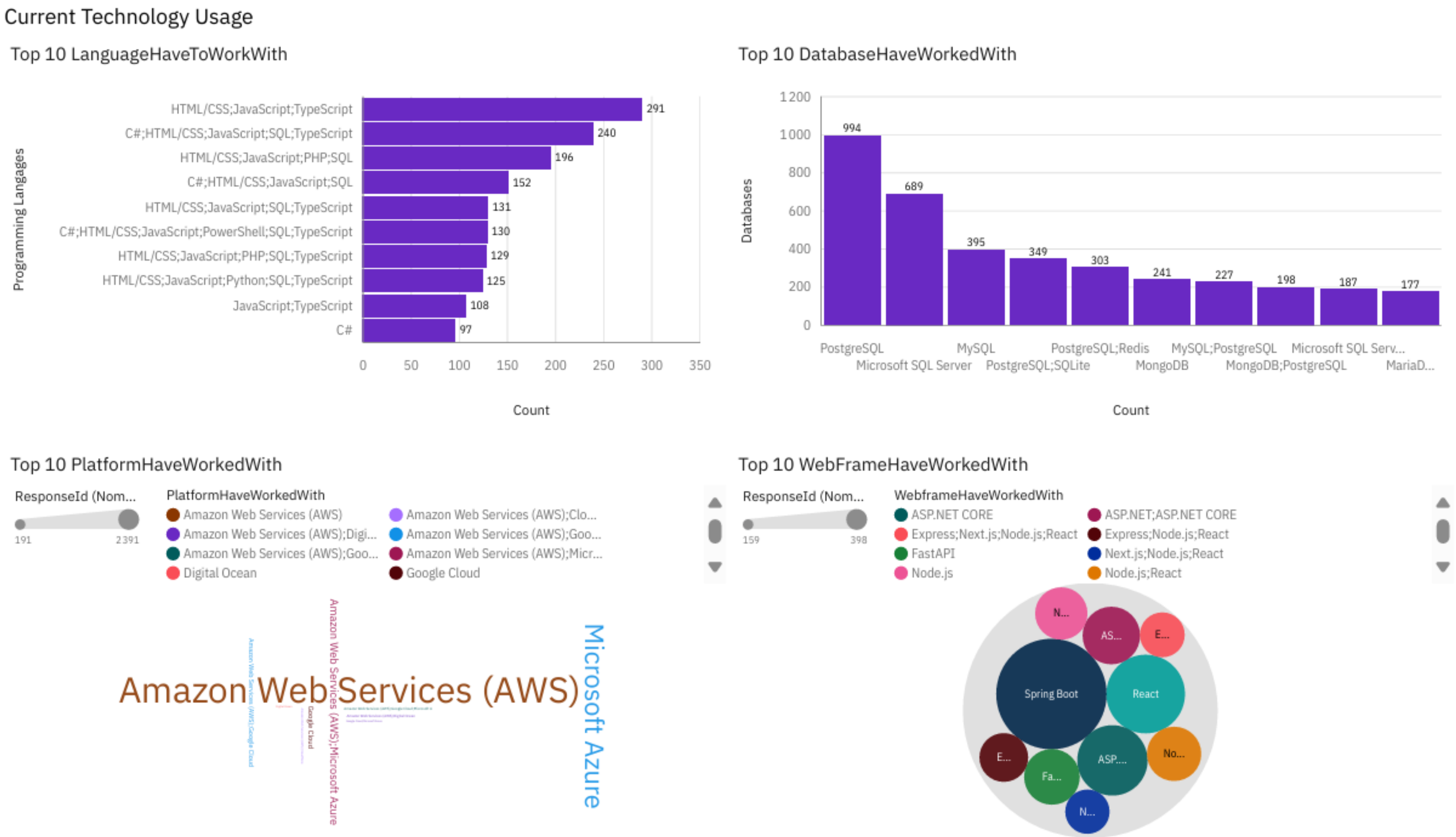
Web Frameworks and Language Interplay Observation: Frameworks heavily influence language use. React (JS), Flask (Python), ASP.NET (C#) are top picks.

-> Framework choices dictate stack adoption in companies. Let’s see the others vizualisations.

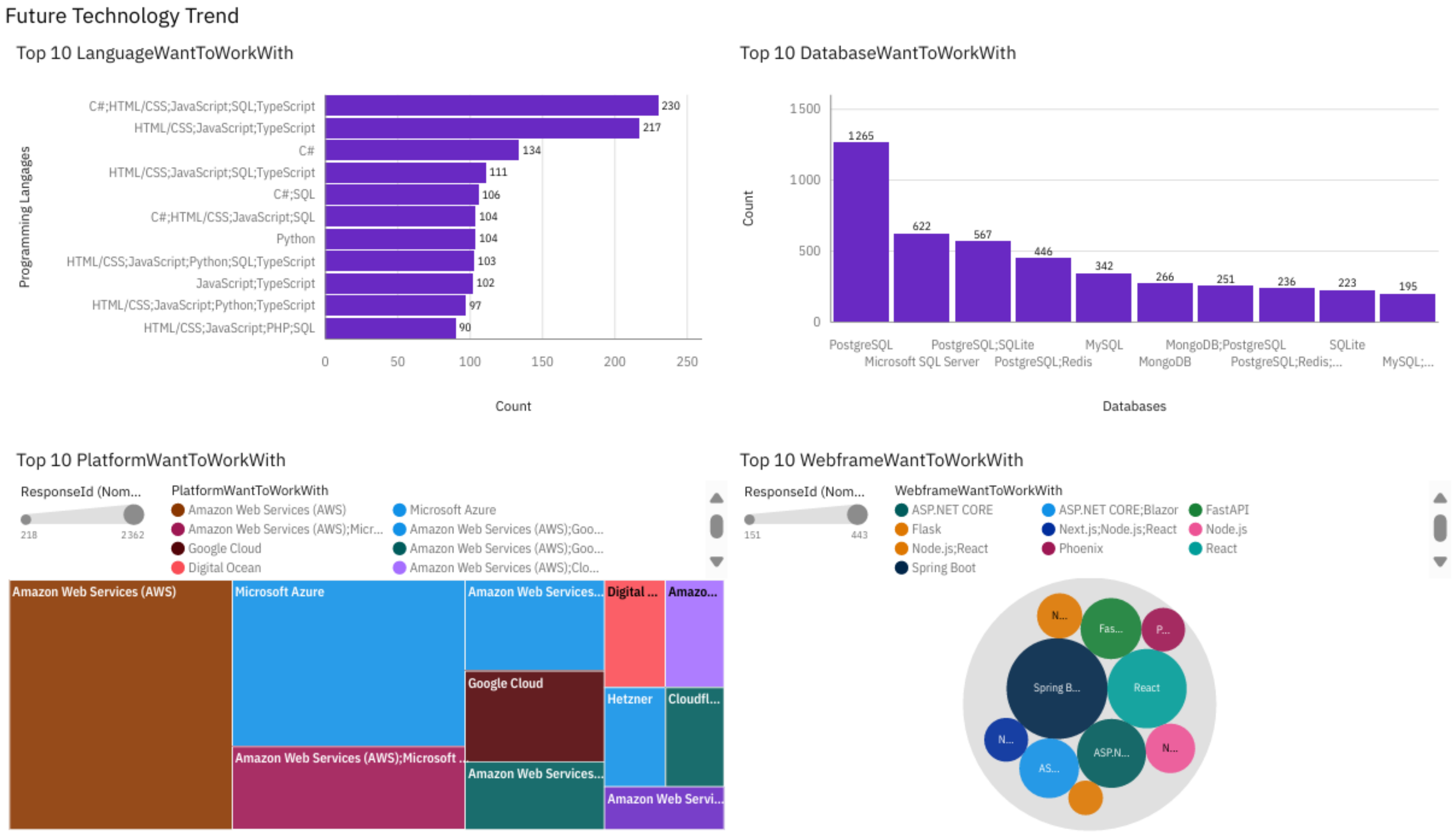
DASHBOARD

The dashboard highlights current preferences for languages like JavaScript and Python, a growing shift toward TypeScript and NoSQL databases, and the influence of web frameworks on language adoption—these insights are explored across three main visualizations: Current Technology Usage, Future Technology Trends, and Demographics.

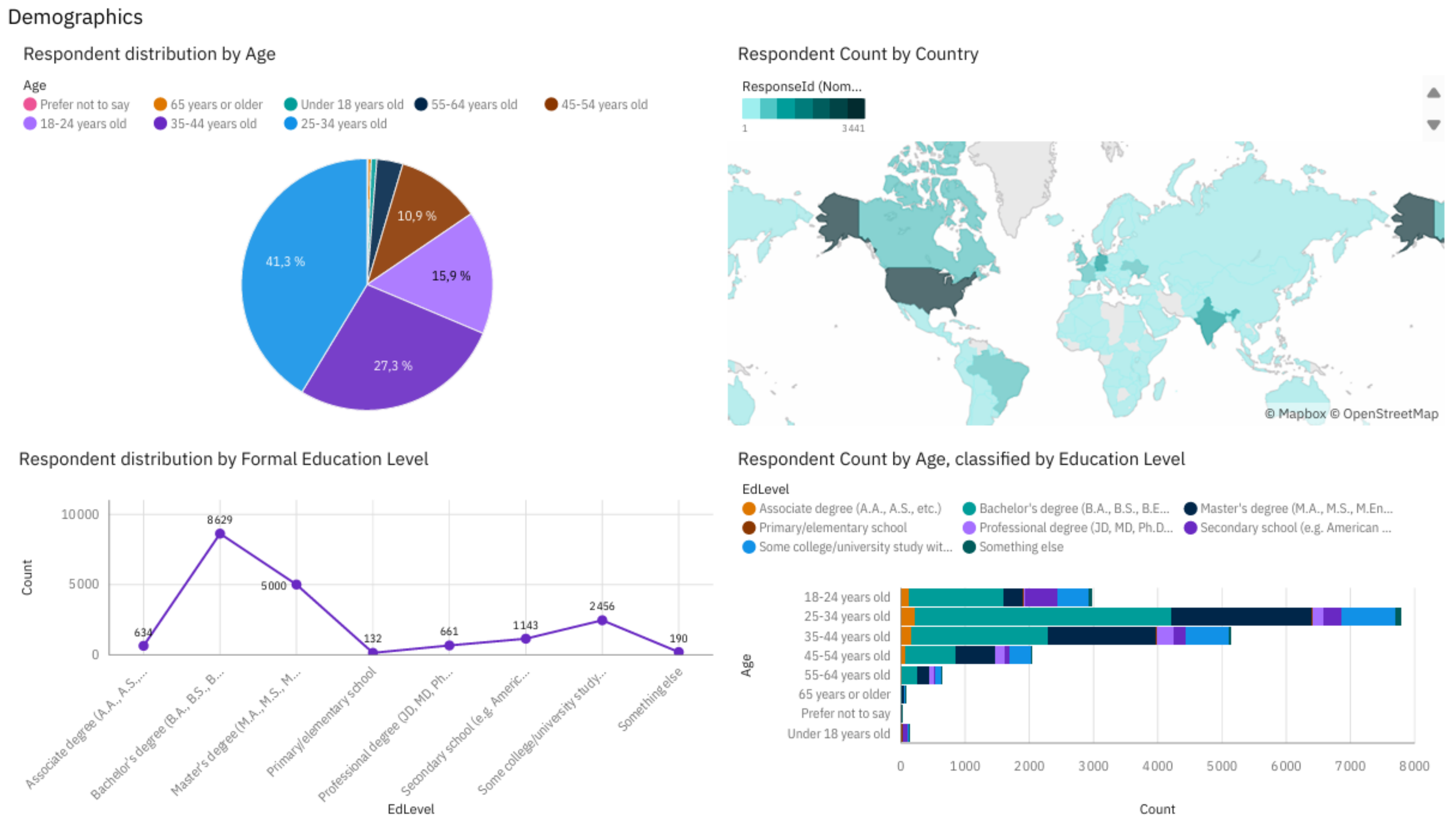
Current Technology Usage



Future Technology Preferences



Demographics (age, education, location)



INSIGHTS FROM THE DASHBOARDS

- Python rising across all age groups and industries
- PostgreSQL preferred in all geographic zones
- Cloud tech (AWS, Azure) dominates professional stack
- Younger professionals (25–34) drive adoption of new tech (e.g., Flask, Firebase)

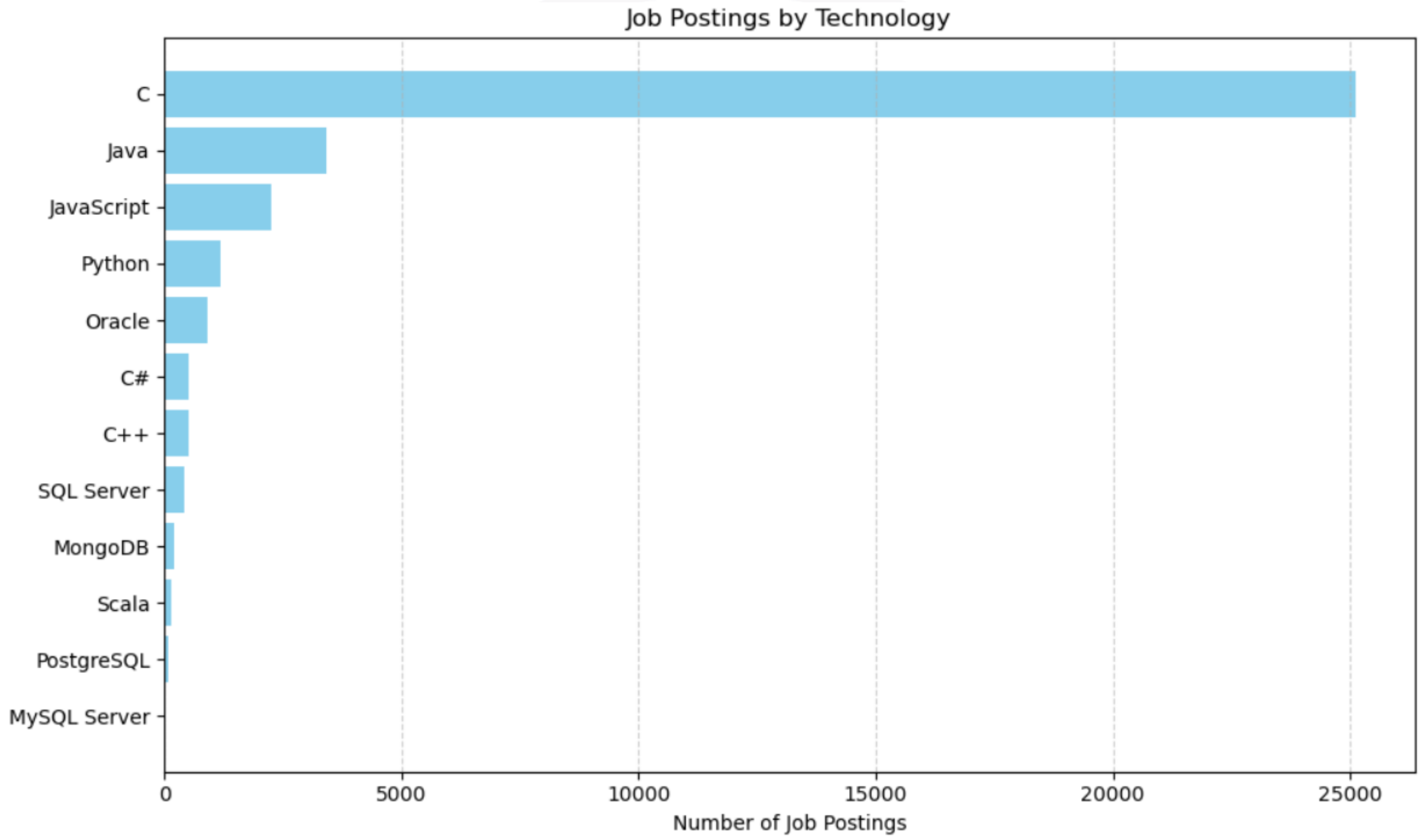
DISCUSSION

Findings reinforce existing research but also highlight emerging shifts not yet mainstream. The rise of No-code and Rustshows how simplicity and speed are becoming critical. The dominance of PostgreSQL reflects a growing preference for open-source, scalable tech. Companies must balance enterprise reliability (Spring Boot, C#) with developer trends (Flask, TypeScript).

CONCLUSION

- Tech stack preferences evolve based on usability, scalability, and developer experience.
- Data confirms that frameworks shape language and DB trends.
- Actionable insight: Update hiring plans and training paths for TypeScript, Python, and PostgreSQL.
- Developers should gain fluency in multi-paradigm stacks.

JOB POSTINGS



POPULAR LANGUAGES